

A PROJECT TITLE

ON

PRODUCT DEMAND FORECASTING USING RIDGE AND LASSO REGRESSION

Project report submitted in partial fulfilment of the requirements for the Course.

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

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1. Abstract:

Accurate demand forecasting is essential for businesses to optimize inventory, minimize wastage, and improve supply chain efficiency. Traditional forecasting approaches often struggle with high-dimensional retail datasets containing correlated features such as price, promotions, seasonality, and historical sales trends, which frequently leads to overfitting when standard linear regression is applied. This project proposes a demand forecasting system for products using Ridge Regression (L2 regularization) and Lasso Regression (L1 regularization), two regularized variants of linear regression designed to address multicollinearity and improve model generalization.

The proposed system uses a publicly available retail sales dataset containing features such as product category, price, discount, seasonal indicators, and past sales volume. The data undergoes preprocessing steps including handling missing values, encoding categorical variables, and feature scaling. Ridge and Lasso regression models are then trained on the processed data, with regularization strength (alpha) tuned using cross-validation to achieve the optimal bias-variance tradeoff. Lasso Regression additionally performs automatic feature selection by shrinking less significant feature coefficients to zero, helping identify the most influential factors driving product demand.

The performance of both models is evaluated and compared using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2) score. A comparative analysis is presented to determine which regularization technique yields better

predictive accuracy and interpretability for this dataset. The expected outcome of this project is a reliable, interpretable forecasting model that can assist retailers and supply chain managers in making informed inventory and procurement decisions, while also demonstrating the practical impact of regularization techniques in real-world regression problems.

2. INTRODUCTION:

Demand forecasting plays a critical role in retail and supply chain management, directly influencing decisions related to inventory planning, procurement, pricing, and staffing. Inaccurate forecasts can lead to overstocking, which increases holding costs, or understocking, which results in lost sales and customer dissatisfaction. With the growth of e-commerce and data-driven retail, machine learning techniques have become increasingly popular for building forecasting models that can capture complex relationships between multiple influencing factors.

Linear Regression is one of the most widely used techniques for demand forecasting due to its simplicity and interpretability. However, when the dataset contains many correlated features, ordinary linear regression tends to overfit the training data, resulting in poor generalization to unseen data. Regularization techniques such as Ridge and Lasso Regression address this limitation by adding a penalty term to the loss function, which constrains the magnitude of the model coefficients.

Ridge Regression applies an L2 penalty that shrinks coefficients towards zero without eliminating them entirely, making it effective when most features contribute to the outcome. Lasso Regression applies an L1 penalty that can shrink some coefficients exactly to zero, effectively performing feature selection. This project explores and compares both techniques for forecasting product demand, aiming to identify which approach provides better accuracy and insight for this problem domain.

3. SOFTWARE REQUIREMENTS:

- Operating System : Windows / Linux / macOS
- Programming Language : Python 3
- IDE / Environment : Jupyter Notebook / Google Colab / VS Code

- Libraries : pandas, numpy, scikit-learn, matplotlib, seaborn
- Dataset Format : CSV (Comma Separated Values)
- Version Control : Git and GitHub